

Discrete Mathematics Assignment 3

Due Date: April 9, 2026

Write clear and concise handwritten solutions. You must also upload a soft copy of it on the moodle page. It is fine to discuss with others, but your solutions must be in your own words that you have fully understood.

1. Find the cycle index polynomial of the group S_5 acting on 2-subsets of vertices. Using this enumerate the number of nonisomorphic graphs on 5 vertices by the number of edges.
2. For n prime count the number of distinguishable n -bead necklaces with beads of k different colors. Suppose you have the additional requirement that no two adjacent beads can have the same color, what is the number of distinguishable beads in that case?
3. Consider a regular tetrahedron in space. Find its three cycle index polynomials (for coloring faces, edges and vertices).
4. Let $c(t) = \sum_n c_n t^n$ be ordinary generating function for counting connected unlabeled graphs. That is, c_n is the number of unlabeled connected graphs (that is, n vertex connected graphs that are not isomorphic to each other). Using the cycle index theorem determine the generating function for counting unlabeled graphs with k components. Prove your result with a clear argument.